Calico Wind Tunnel

Project dev notes

Figure 1: A full view of the wind tunnel during an experiment for boundary layer height.

Project aim. To build an affordable, portable wind tunnel that high-school students can manufacture in an ordinary workshop and use for genuine aerodynamic experiments, then publish the design so other students can build their own.

1. How the project began

I first encountered a wind tunnel during the Frontiers summer program at Worcester Polytechnic Institute (WPI), between Grades 9 and 10. I learned how to operate a research wind tunnel and use computational fluid dynamics (CFD) as part of the process. That experience helped spark my interest in aerospace engineering.

Figure 2: My group and I operating the wind tunnel.

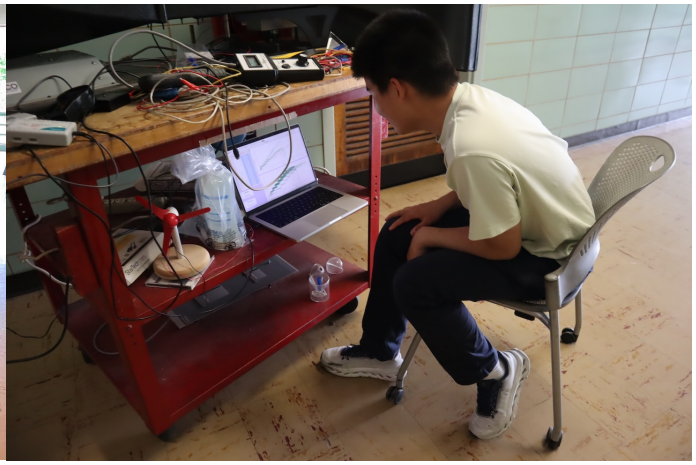


Figure 3: Myself independently recording data from wind tunnel

I later began building my own aircraft. Because of Beijing's restrictions on drone flying, I chose to work on Remote Control (RC) gliders rather than conventional multirotor drones. As I developed the gliders, I used CFD to investigate a possible flying-wing design. However, I found that CFD could be unintuitive and expensive to use, especially when I needed to run many simulations at different speeds and angles of attack. I eventually ran out of my free simulation credits and returned to a more conventional glider design.

In late 2025, Beijing tightened its drone regulations further, so I had to pause my flying projects and wait until I go to college elsewhere to resume them. That led me to ask what I could still build and investigate. A wind tunnel offered a way to continue exploring aerodynamics and test physical designs, even when I could not fly them outdoors.

2. Three iterations: from a demonstration to a research tool

Version 1: a very low-cost prototype

In the latter half of Grade 10, I designed a tiny wind tunnel using a PC fan, 3D-printed parts, and acrylic. It cost less than US$15 and worked well for its size and price. The aerospace club I was teaching also enjoyed using it. This prototype showed me that a wind tunnel could be made inexpensively, but it was only a starting point.

Figure 4, 5, 6: Version 1 being used in the classroom of The Aerospace Club. Image courtesy: Michael Shen

Version 2: learning from an open-source design

My second version was based on an open-source wind-tunnel design made largely with 3D-printed parts. It included several useful features and taught me a lot about the construction process, but its test section was too small for many of the experiments I wanted to conduct. I assembled it with the aerospace club, which also revealed a practical problem: students without much engineering or workshop experience found it difficult to install.

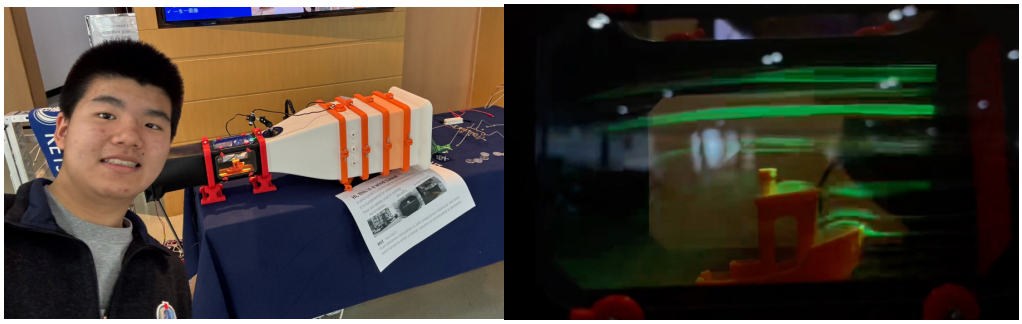


Figure 7, 8: Version 2 Wind Tunnel on display at a tech fair in our school and testing footage.

Version 3: the current project

I then began the third version, which has become a capstone to my high-school engineering and workshop experience. My aim is to make a wind tunnel that I and other students can use for genuine research, rather than only for demonstrations. I also want to open-source the design so that other high-school students can build their own.

3. Design priorities

I designed the current version around materials and fabrication methods that are available in many school workshops, while minimizing the number of parts that need to be specially ordered. The main materials are plywood, aluminum extrusions, polycarbonate (PC) sheets, and PLA parts made on a 3D printer. I planned the components so that they could be manufactured using common workshop tools such as saws, a laser cutter, and a 3D printer. The plywood components can also be CNC-routed to speed up production.

I also made the CAD files easy to navigate and parametric, so another builder can change key dimensions to suit their needs. This matters to me because an open-source design should be adaptable, not simply a fixed set of files that other people have to copy exactly.

Portability became another design requirement after a teacher saw my second version and asked whether I could make a wind tunnel for science-class demonstrations. I therefore designed the current tunnel to be stowed in a compact configuration and moved around school, including through an elevator.

4. Building the current version

Planning and procurement

I began the current version's CAD design in mid-October, focusing first on the overall layout and on making the files clear and easy to navigate. I also checked that the parametric CAD model worked as intended. I worked on the first physical components in January alongside other projects. The pace increased after I sent our FRC robot to competition in mid-March: I prepared a detailed purchasing list covering the parts I needed and began assembling the test section.

Figure 9: CAD Main Assembly, Annotated.

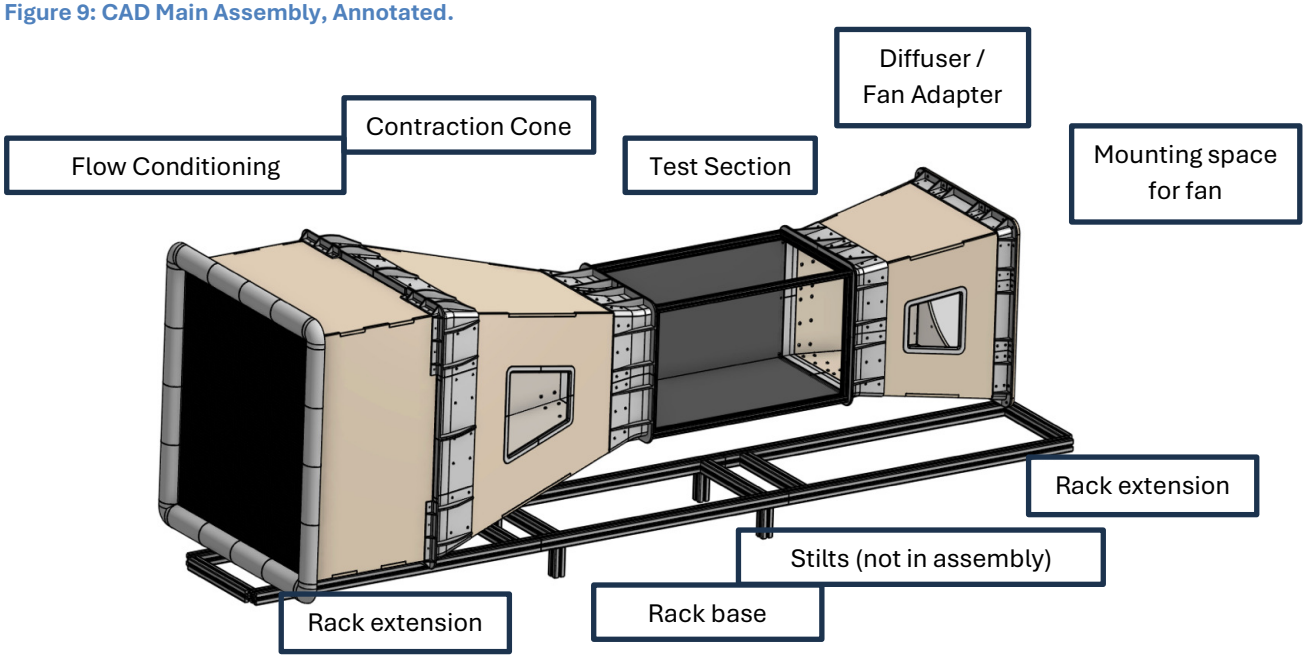


Figure 10, 11, 12: Exploring the general configuration of the wind tunnel.

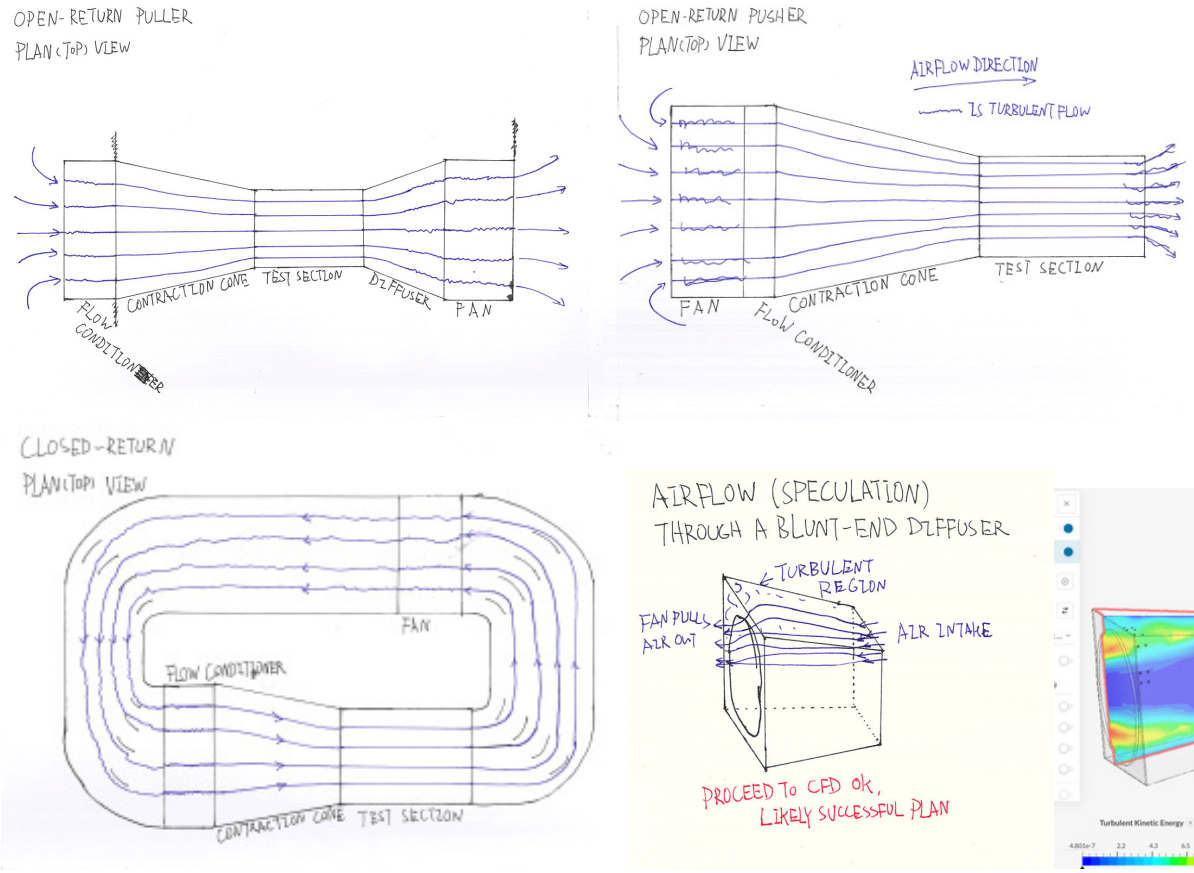


Figure 13, 15, 16, 17: Part-by-part design sketches.

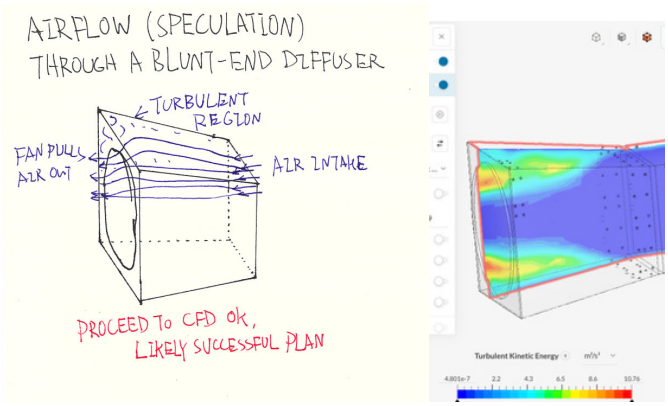
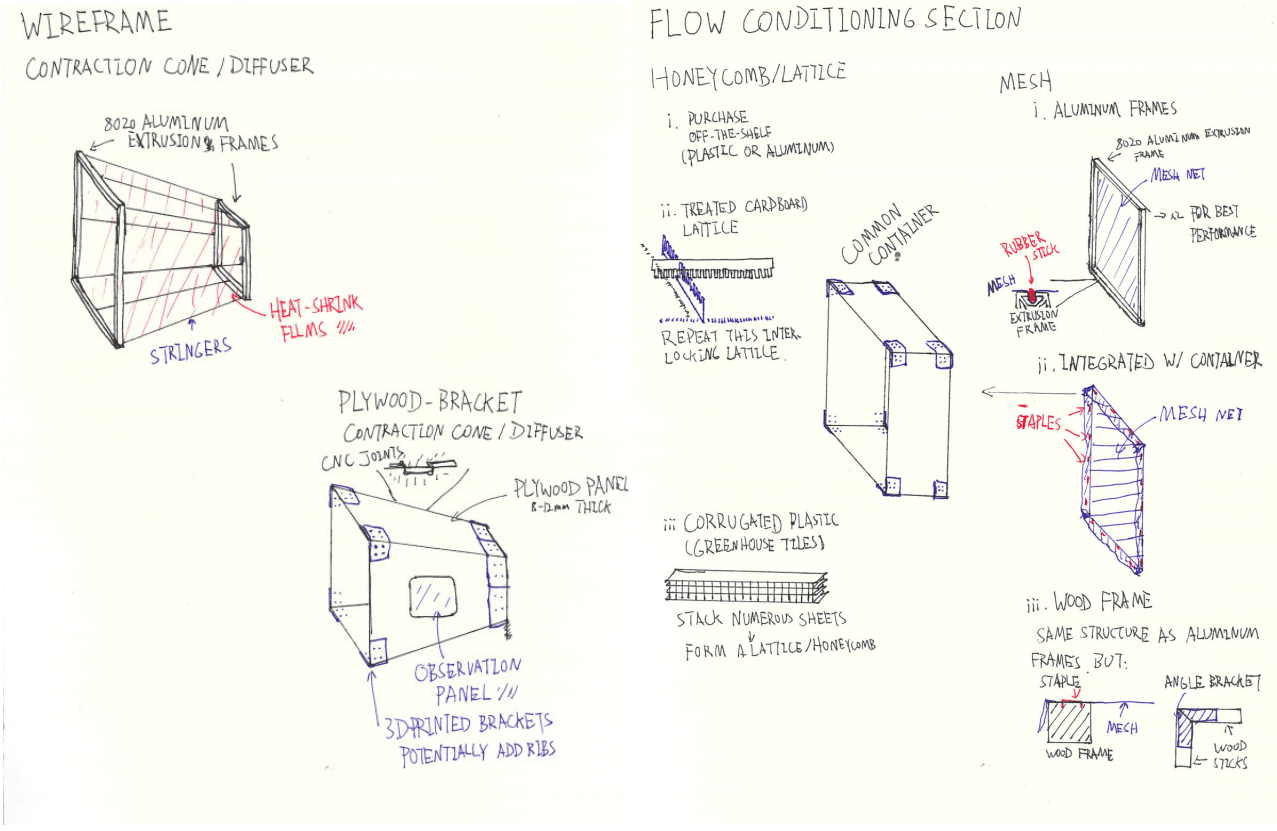
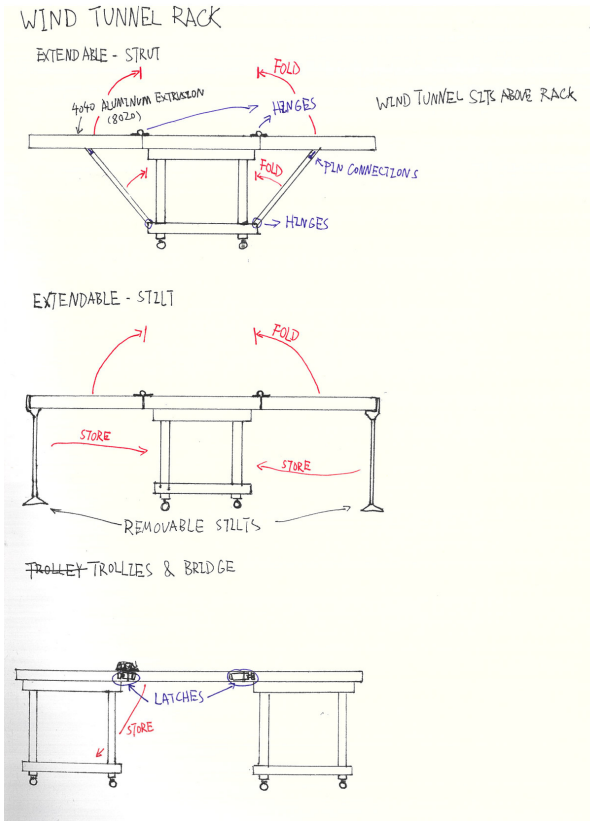


Figure 14: CFD validation of the Figure 13 design





Item Name	Specifications	Quantity	Unit Cost	Total Price	Link
Axial Fan	岗位式5-4/单相220V/30公分引线无插	1	340	340	https://item.taobao.com/item.htr
VFD	2.2kw单相220进单相220出	1	370	370	https://item.taobao.com/item.htr
2020 Al Extrusion	Per meter price	5	12.3	61.5	https://item.taobao.com/item.htr
Extrusion Hinge	1个20银色合页（含螺丝螺母）	4	10	40	https://item.taobao.com/item.htr
Extrusion Connectors	欧标20型M5	8	1.5	12	https://item.taobao.com/item.htr
Wing Bolt	M5*8-10个	5	2.41	12.05	https://item.taobao.com/item.htr
Grant total				835.55	
The Above is already Purchased					
Need to Purchase					
Item Name	Specifications	Quantity	Unit Cost	Total Price	Link
Trolley	100*60*80 cm, length*width*height	1	205.47	205.47	https://item.taobao.com/item.htr
Quick-Release Clamp	not: A字夹-4寸/10个	2	18.9	37.8	https://detail.tmall.com/item.htm
	Flat head screw (internal hex), M5*20mm	15	4.32	64.8	https://detail.tmall.com/item.htm
	Lock Nut, M5	8	4.44	35.52	https://detail.tmall.com/item.htm
	Washer, M5*10*0.5mm	1	3.08	3.08	https://detail.tmall.com/item.htm
	4040 Straight Connector, 欧标40型-M6	4	4.3	17.2	https://detail.tmall.com/item.htm
	M5 4040 T-nut, 40型-M5 (镀镍/20个)	2	3.25	6.5	https://detail.tmall.com/item.htm
	M5 2020 T-nut, 20型-M5 (镀镍/50个)	1	3.57	3.57	https://detail.tmall.com/item.htm
	4040 Extrusion angle bracket,4040角码	4	1.9	1.9	https://detail.tmall.com/item.htm
	2020 Extrusion 90° insert, 欧标2020-M	1	11.25	11.25	https://detail.tmall.com/item.htm
	2020 Extrusion Angle Bracket, 2020角码		1.15	0	https://detail.tmall.com/item.htm
Fasteners	4040 extrusion hinges, 1个40银色合页（含螺丝螺母）	2	12	24	https://item.taobao.com/item.htr
PLA Filaments	6kg pack, no reel (there is a discount). Overbought because keytec still need it	2	312	624	https://detail.tmall.com/item.htm
Fillers	150ml一支【赠安装四件套】	1	10.9	10.9	https://detail.tmall.com/item.htm
Sealing Rubber Strips	Φ3mm（10米价）	2	3.5	7	https://detail.tmall.com/item.htm
Grayboards	A1 1mm（双面灰板纸）10张	2	66.5	133	https://item.taobao.com/item.htr
Varnish	油性光油【亮光清漆-透明色】350ml*	1	8	8	https://detail.tmall.com/item.htm
Hot Wire Anemometer	TBD			0	
Laser	TBD			0	
Total: (expect about 5-10% off for the products. Now everything is on sale)				1193.99	

Filament Calculation				
Assembly	Part Name	Part Count	Weight (g)	Total (g)
A - Flow Conditioning	Clamping Connector	4	71	298.2
	Bellmouth Corner	4	64	268.8
	Bellmouth Mid	8	161	1352.4
B - Contraction	Bracket In Corner	4	332	1394.4
	Bracket in Mid	8	165	1386
C - Test Section	Bracket Out	4	358	1503.6
	Estimate	1	300	315
D - Diffuser	Bracket Out Corner	4	98	411.6
	Bracket Out Mid	4	67	281.4
	Bracket In	4	270	1134
Additional	Estimate	1	500	525
				8870.4

Figure 18: Main Purchasing List screenshot.

Throughout the design, I referenced online guidebooks about fluid dynamics (specifically flow separation, inverse pressure gradient, and boundary layer theory) and wind tunnel construction to be generally informed about how specific components will disturb the airflow. For example, I chose flathead bolts instead of cup hear bolts, because the former one is well within the slow-flowing boundary layer of air, and the latter one is on the edge of it, having a larger chance of disturbing the general airflow.

Trolley and extendable support frame

As parts arrived, I assembled a trolley to support the wind tunnel. The countertop was structurally unsound, so a strengthening panel was added to further ensure structural integrity.

Figure 19: The assembled trolley and plywood reinforcement sheet on the side.

I then built the extendable rack. It failed its first test, so I added a strengthener and checked the frame using laser levels. It passed the initial static test, but the assembled structure later showed that the rack still sagged about 8mm under the full weight of the tunnel. I added removable support stilts at the sides, which resolved the problem. This was a useful reminder that a design can pass an early test and still need revision under real operating conditions.

Figure 20, 21: The first version sags under own weight.

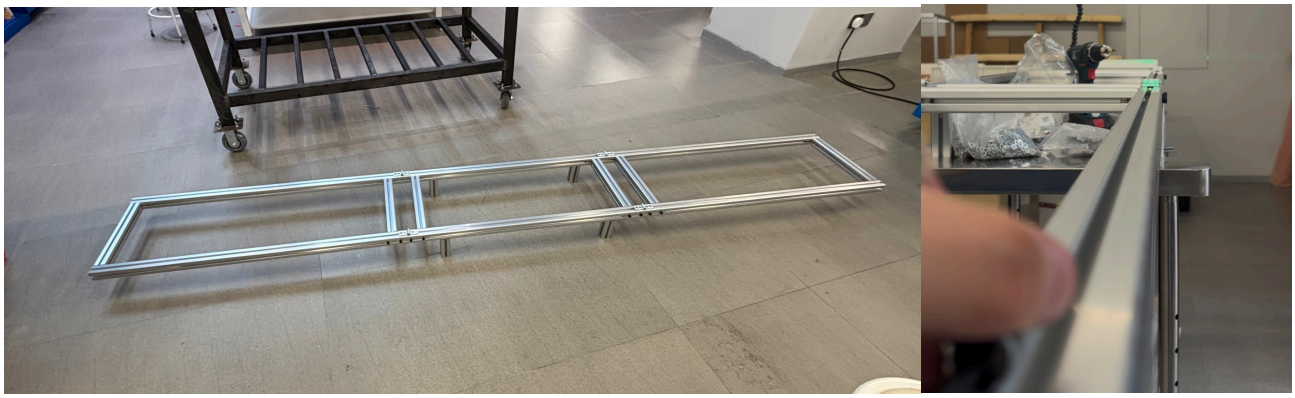


Figure 22, 23: The second version remains intact under its own weight, checked using laser levels.

Figure 24, 25: However, the 2nd version is not sufficiently strong. Stilts seen in Fig. 25 solves this problem entirely.

Contraction cone, diffuser, and stilts

I manufactured and assembled the contraction cone and diffuser. After validating sample pieces, I had the plywood sheets CNC-routed because this greatly accelerated production. I had also designed the components so they could be cut by hand or with a laser cutter where CNC routing was unavailable. I then joined the plywood panels and 3D-printed parts to form the structure, using sealant along the joints to improve airtightness. This assembly was fairly easy, thanks to previous experiences in making wooden structures in theater backstaging.

The stilts are assembled by a group of friends interested in this project. Their work helped me greatly on this part of the project. After the project is finished, all contributors' names would be laser-engraved on the plywood in a location of their choice.

Figure 26, 27: Assembly of contraction cone and diffuser.

Figure 28: Some old planks are recycled for spacers in the process, in hope to reduce environmental impact.

Figure 29: A fellow friend assembling the stilts used to support the main structure

Figure 30: A selfie taken during construction.

Flow-conditioning section

The final major component was the flow-conditioning section. It contains a lattice commonly called a honeycomb, although the cells are not necessarily hexagonal, followed by two layers of mesh. The honeycomb helps straighten the incoming airflow, while the mesh helps reduce smaller flow disturbances.

Initially, the structure was made out of acrylic-sprayed cardboard, but the assembly encountered too much friction, so I had to use the second plan: a corrugated plastic solar tile instead. This droves up the cost by 400 CNY (≈55 USD). The section currently performs most of its intended functions, but it is difficult to move, so its structure still needs improvement.

Figure 31, 32: First attempt of honeycomb with cardboard is unsuccessful. The final version is corrugated PC.

Figure 33, 34: Interior views of the flow conditioner.

5. Testing and using the wind tunnel

I carried out several kinds of tests to check the tunnel's construction and performance:

- **Sealing tests.** I used a smoke machine to check for leaks. Smoke being drawn into the tunnel at a joint indicated that the seal was not adequate. I used what I learned from these tests to improve the design so that the structure was more reliable and the seals were easier to repair.

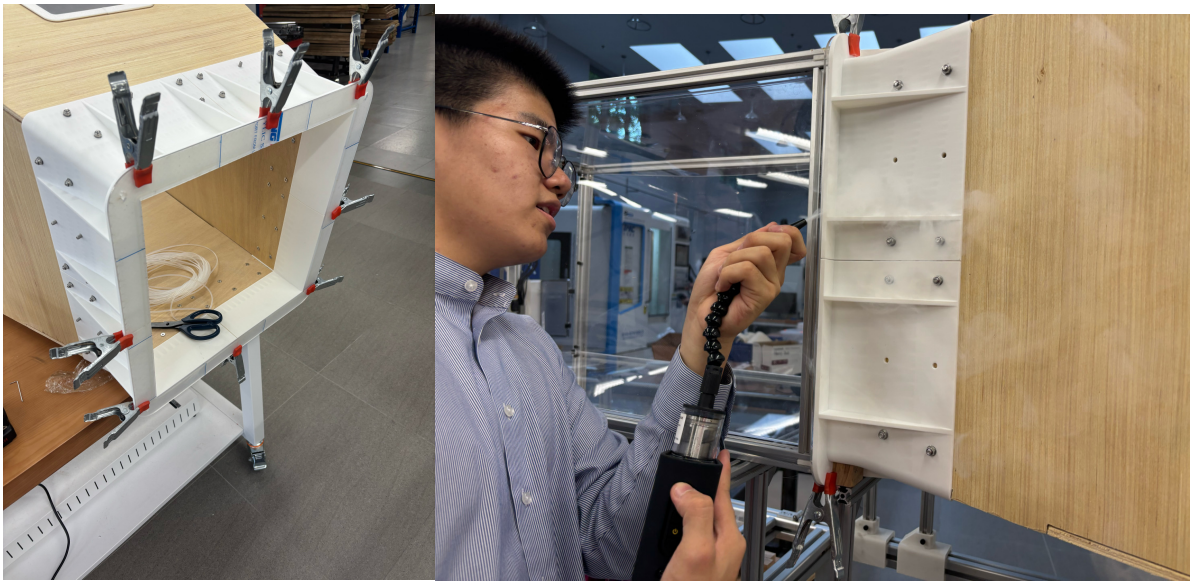


Figure 35: Testing if clamps are sufficient enough to seal the interface between test section and diffuser.

Figure 36, 37: Testing of these seals shows that they are sealed, with negligible leakage.

Figure 38: Testing shows significant leakage near fan area, fixed later and confirmed by same testing method.

- **Structural tests.** I used laser levels to check alignment and added loads to stress-test the extendable racks. Before installation, I loaded the racks with heavy weights beyond what I expected them to carry in normal use and shook the structure. It held up during these tests. I also included structural redundancies so that a single component failure would be less likely to create a safety risk.

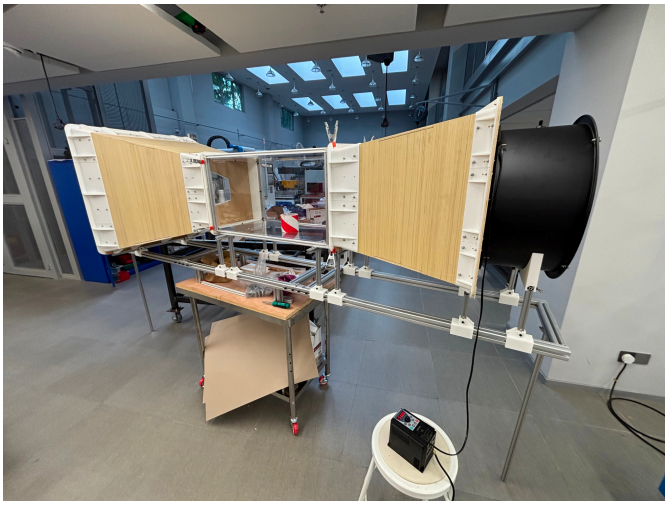
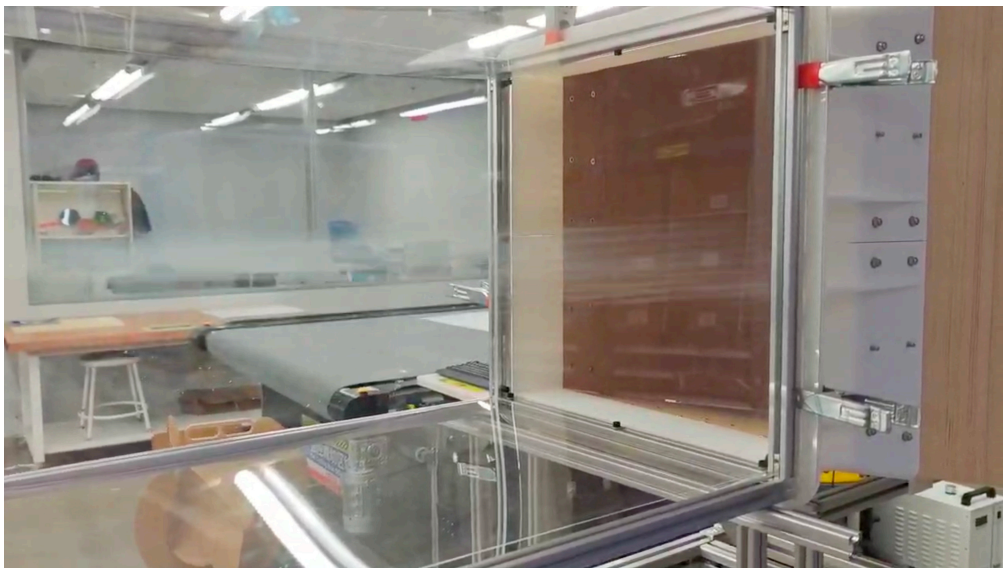
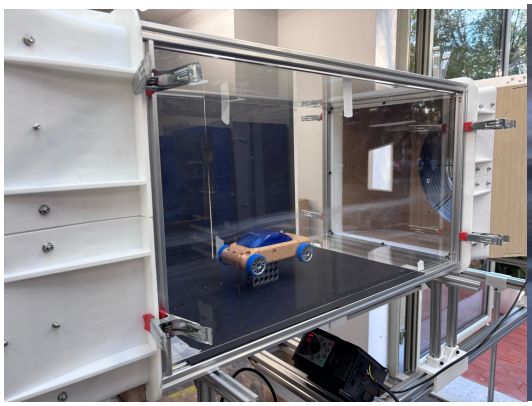


Figure 39: The full-load full-power structure test. Vibration exists, but are within safety limits.

- **Flow tests.** I used smoke and a hot-wire anemometer to examine the airflow. These tests indicated that velocity was reasonably consistent and that the flow was broadly laminar, although further improvements and measurements are still needed.





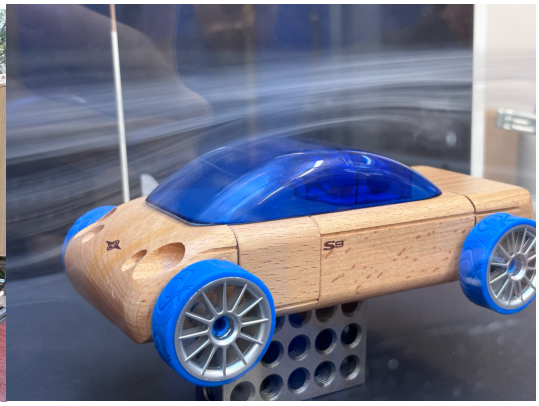


Figure 40, 41, 42: Preliminary testing of the flow quality using nearby objects. Shows promising results.

After the build, I decided to name the wind tunnel Calico, inspired by a teacher supervising my project in the workshop. The name stems from the three main colors of the wind tunnel: white PLA, plywood, and black cardboard and fan.

Figure 43: An example of calico cat. Very common variety in China.

A first research application: my Physics Extended Essay

I have completed my Physics Extended Essay experiment using the wind tunnel. My investigation examines the boundary layer over a flat plate using the Blasius solution—in other words, how airflow slows as it approaches a surface. The results look promising and mostly agree with the theoretical prediction, though there are limitations to address. For example, the whole tunnel resonates at around 9 m/s, so I had to omit the trial at that speed. Identifying this limitation has helped me see where the apparatus needs further refinement before it can support a wider range of experiments.

Figure 44: Me carrying out the boundary layer flow experiment. Results show that the wind tunnel produces near-laminar flow, as the flow behavior matches the Blasius solution quite closely, with minimal outliers.

6. Collaboration and wider use

Throughout construction, I invited other people to help whenever they had time. This gave them hands-on experience of contributing to a substantial engineering project, and it gave me a chance to observe which steps other builders found difficult. Their experience helped me think more carefully about how the design could be made easier for other students to assemble.

The wind tunnel is also used to demonstrate what our workshop and students can make. We have placed a range of demonstration objects inside it. In addition, after visiting our school, a design teacher from another local school proposed a collaboration involving one of their students and me; that collaboration is now underway.

Several students and teams are also preparing to test their designs once the tunnel is ready for broader use. These include an ornithopter, a bicycle component, and aerodynamic parts from an F1 racing club. This is important to me because it means the project could become shared equipment that supports other students' investigations, rather than remaining a personal build.

7. Next steps and reflection

My next steps are to improve the flow-conditioning section so that it is easier to handle, investigate the resonance around 9 m/s, and manufacture the free-floating sting system I have already designed. The sting system will allow model tests and lift measurements of components. I plan to manufacture it after I finish the early-application round.

This project has taught me how to manage a long-term build, plan and order components, coordinate help from others, test a structure, and revise a design when the first solution does not work. I have also learned new fabrication techniques and gained a clearer understanding of what I and our school workshop can accomplish. Most importantly, I am building a tool that could make practical aerodynamic research more accessible to other high-school students.

AI Use Disclosure

I used ChatGPT (OpenAI) to help reorganize these project notes, improve the clarity of the writing, and correct grammatical errors. The original project notes and the entirety of the underlying engineering work, including the modelling, design decisions, construction process, testing, and personal reflections, are my own. I reviewed the revised text to ensure that it accurately represents my work and take full responsibility for the final submission.

The images, sketches, and other visuals, unless stated otherwise, are all created by Derek Han, the main developer of this project and the writer of this dev log. All people with their face present in the images gave explicit consent for the use of these images in the documentation of this project.